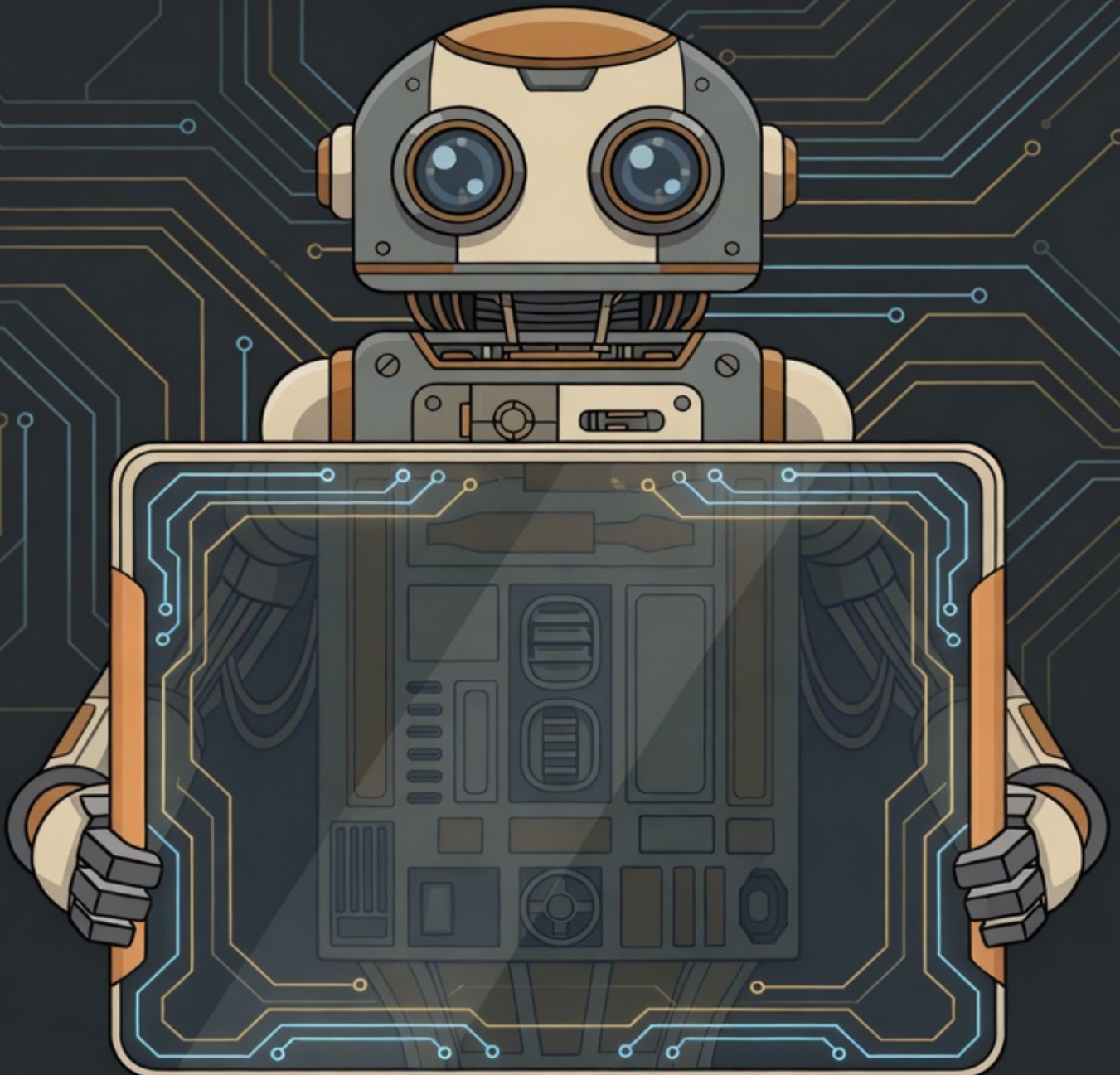


ARLIZ

Go Beyond Arrays, Memory, and Logic —
A Journey from Foundations to Frontiers

Mahdi (m-mdy-m)



ARLIZ

ARRAYS • REASONING • LOGIC • IDENTITY • ZERO

*"From ancient counting stones to quantum algorithms—
every data structure tells the story of human ingenuity."*

LIVING FIRST EDITION

Updated: June 13, 2026

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A Living Architecture of Computing

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Preface

Why Would Anyone Learn This Stuff?

The Name and Its Meaning

What This Book Represents

Arliz is not a gentle introduction to programming, nor is it a purely theoretical treatment of data structures. Instead, it represents something more focused and, I believe, more valuable: a comprehensive technical exploration of the most fundamental data structure in computing, examined from every relevant angle.

The structure reflects a deliberate progression through increasingly sophisticated understanding:

- **Data Representation** — How information is encoded in digital systems, from number systems to character encoding
- **Computer Architecture and Logic** — The hardware foundations that determine how arrays actually work
- **Array Odyssey** — Deep exploration of array implementation, behavior, and optimization

Mahdi
2026

“The purpose of abstraction is not to be vague, but to create a new semantic level in which one can be absolutely precise.”

— EDSGER W. DIJKSTRA

Introduction

ARRAYS ARE EVERYWHERE. Every image you view, every text you read, every game you play, every database query you execute—arrays underlie them all. Yet despite their ubiquity, arrays remain poorly understood by most programmers. They are treated as primitive constructs, learned hastily and used mechanically, their true nature obscured by layers of abstraction.

This book exists to remedy that situation.

What This Book Is

Arliz is a focused, technical exploration of the most fundamental data structure in computing: the array. Unlike comprehensive data structures textbooks that survey many structures superficially, we examine arrays with unprecedented depth—from their hardware representation in memory through their implementation in programming languages to their role in advanced algorithmic techniques.

This is not a book for casual reading. It demands engagement, rewards persistence, and assumes substantial mathematical and computational maturity. If you seek quick tutorials or surface-level explanations, you will find this book frustrating. If you seek deep, rigorous understanding of how arrays actually work and why they behave as they do, you have found the right resource.

The Architecture of This Work

The book progresses through three major parts, each building on the foundations established by its predecessors:

Part I: Data Representation

We begin with the fundamental question: how is information encoded in digital systems? From binary representation through character encoding, from integer formats to floating-point arithmetic, we establish the representational foundations that determine how array elements are actually stored and manipulated at the machine level.

Part II: Computer Architecture & Logic

Arrays do not exist in abstract space—they live in physical hardware with specific characteristics and constraints. We examine logic gates, processor architecture, memory hierarchies, and cache behavior. Understanding these hardware realities is essential for writing array code that performs well on real systems.

Part III: Array Odyssey

This is the heart of the book. We explore arrays from every angle: their mathematical properties, their implementation in memory, their performance characteristics, their optimization techniques. We examine one-dimensional arrays, multidimensional arrays, sparse arrays, and specialized array variants. We analyze cache behavior, memory alignment, and hardware-level optimization strategies.

Prerequisites

This book assumes substantial preparation:

Mathematical Maturity: You should be comfortable with discrete mathematics, linear algebra, and basic mathematical analysis. Specifically, you should have completed (or be comfortable with) the material in *Mathesis: The Mathematical Foundations of Computing*.

Algorithmic Analysis: You should understand asymptotic notation, recurrence relations, and basic complexity analysis. The material in *The Art of Algorithmic Analysis* provides the necessary foundations.

Without these prerequisites, you will struggle with substantial portions of this book. With them, you are prepared for a deep, rewarding exploration of arrays.

The Living Nature of This Work

Like all my books, *Arliz* evolves continuously. As I discover better explanations, identify errors, recognize new connections, or encounter new implementation techniques, the book improves. Your engagement—through corrections, suggestions, implementations, and questions—contributes to this evolution.

Check the GitHub repository regularly for updates. The version you read today will be superseded by improved versions tomorrow. This is not a weakness but a strength—the book remains current, accurate, and relevant. I hope this book serves you well—whether you are building your first serious data structures, optimizing performance-critical systems, or simply satisfying intellectual curiosity about how arrays actually work.

A Note on Difficulty

This book is challenging by design. Arrays may seem simple on the surface, but their full understanding requires grappling with hardware architecture, memory systems, compiler optimizations, and algorithmic analysis. Some sections will require sustained effort to master.

This difficulty is intentional. Genuine understanding is never cheap—it demands intellectual investment, persistent effort, and willingness to struggle with complex concepts. If you find sections difficult, that is normal. Persist. Work through examples. Implement the techniques. Understanding will come through sustained engagement.

Begin

Welcome to **Arliz**. Let us explore arrays with the depth and rigor they deserve.

“Premature optimization is the root of all evil.”

— DONALD E. KNUTH

Part I

Data Representation

***B**EFORE WE can understand how arrays store elements, we must first understand how computers represent information itself. Every piece of data—numbers, text, images, instructions—exists as patterns of bits. This part explores the foundational question: how do we encode meaning into binary?*

“The choice of representation is often more important than the choice of algorithm.”

— DONALD E. KNUTH

Chapter 1

The Nature of Data

1.1 What is Data

1.2 Data in Information Theory

1.3 The Relationship Between Data and Representation

Chapter 2

The Philosophy of Representation

2.1 What Is Representation

2.1.1 Why Representation Matters

2.1.2 Representation In Information Theory

Chapter 3

Voltage to Bits

How transistors encode binary states, voltage levels, noise margins, signal integrity, why binary won over ternary.

Chapter 4

Binary Representation and Boolean Algebra

Two-state logic, Boolean operations, truth tables, De Morgan's laws, bit as fundamental unit.

Chapter 5

Number Systems and Bases

Positional notation, decimal, binary, octal, hexadecimal. Base conversion algorithms. Historical development. Why different bases matter for different purposes.

Chapter 6

Integer Representation: Unsigned

Binary positional notation, range limitations, conversion algorithms, hexadecimal convenience.

Chapter 7

Signed Integer Representation

Sign-magnitude, one's complement, two's complement (why it won), arithmetic operations, overflow detection.

Chapter 8

Integer Overflow and Wraparound Behavior

Undefined behavior, modular arithmetic, detecting overflow, saturating arithmetic, language-specific behaviors.

Chapter 9

Fixed-Point Representation

Q-format notation, scaling factors, precision-range tradeoffs, embedded systems applications, fixed-point arithmetic.

Chapter 10

Floating-Point Representation: IEEE 754

Sign, exponent, mantissa encoding, normalized and denormalized numbers, single vs. double precision.

Chapter 11

Special Floating-Point Values

Infinity (positive and negative), NaN (quiet and signaling), signed zero, subnormal numbers.

Chapter 12

Floating-Point Arithmetic Operations

Addition, multiplication, division, rounding modes (round to nearest, toward zero, toward), fused multiply-add.

Chapter 13

Floating-Point Error Analysis

Precision loss, catastrophic cancellation, machine epsilon, relative error, Kahan summation algorithm.

Chapter 14

Decimal Floating-Point Representation

IEEE 754-2008 decimal formats, financial computing requirements, densely packed decimal.

Chapter 15

Extended Precision and Arbitrary Precision

80-bit extended precision, quadruple precision (128-bit), arbitrary precision libraries (GMP, MPFR).

Chapter 16

Character Encoding: ASCII and Extensions

7-bit ASCII, extended ASCII variants, code pages, limitations for internationalization.

Chapter 17

Unicode: Universal Character Encoding

Code points, planes, combining characters, grapheme clusters, normalization forms (NFC, NFD, NFKC, NFKD).

Chapter 18

Unicode Transformation Formats

UTF-8 (variable length, backward compatible), UTF-16 (surrogate pairs), UTF-32 (fixed length), BOM issues.

Chapter 19

Text Processing Complexities

Grapheme vs. code point counting, case folding, locale-dependent operations, text segmentation.

Chapter 20

Endianness: Byte Ordering

Big-endian vs. little-endian, historical reasons, network byte order, bi-endian systems, byte swapping.

Chapter 21

Cross-Platform Data Exchange

Serialization formats, portable binary formats, protocol design considerations.

Chapter 22

Bitwise Operations Fundamentals

AND, OR, XOR, NOT operations, truth tables, bit manipulation primitives.

Chapter 23

Bit Shifting and Rotation

Logical shift, arithmetic shift, rotate left/right, applications to multiplication/division by powers of 2.

Chapter 24

Bit Manipulation Techniques

Setting/clearing/toggling bits, bit masks, extracting bit fields, counting set bits (population count).

Chapter 25

Bit Packing and Flags

Packing multiple boolean flags, bitfields in structs, union tricks, bit arrays.

Chapter 26

Advanced Bit Hacking

Finding rightmost set bit, isolating bit patterns, bit reversal, Gray code conversion.

Chapter 27

Data Alignment Fundamentals

Natural alignment, alignment requirements by type, misalignment penalties.

Chapter 28

Structure Padding and Layout

Compiler padding rules, struct member ordering, packing pragmas, cache line awareness.

Chapter 29

Alignment Optimization Techniques

Manual padding, alignment attributes, reordering for cache efficiency.

Chapter 30

Color Representation Models

RGB, RGBA, HSV, HSL, CMYK, YUV, color spaces, gamma correction.

Chapter 31

Pixel Formats and Bit Depth

1-bit, 8-bit indexed, 16-bit (RGB565), 24-bit, 32-bit (with alpha), HDR formats.

Chapter 32

Image Compression Fundamentals

Lossless (PNG, GIF) vs. lossy (JPEG), compression ratios, quality tradeoffs.

Chapter 33

Video Encoding Principles

Frame types (I, P, B), motion compensation, codecs (H.264, H.265, VP9, AV1).

Chapter 34

Audio Representation: Sampling Theory

Nyquist theorem, sample rate, bit depth, quantization noise, aliasing.

Chapter 35

Audio Encoding Formats

PCM (uncompressed), FLAC (lossless), MP3, AAC, Opus (lossy), perceptual coding.

Chapter 36

Signal Processing Representations

Time domain, frequency domain, spectrograms, wavelet transforms.

Chapter 37

Pointer Representation

How pointers are stored in memory, pointer size (32-bit vs. 64-bit), pointer tagging.

Chapter 38

Pointer Arithmetic and Address Calculation

Array element addressing, struct member offsets, pointer subtraction.

Chapter 39

Special Pointer Values

Null pointers, wild pointers, dangling pointers, pointer alignment requirements.

Chapter 40

Error Detection Codes

Parity bits (even/odd), checksums, CRC algorithms, hash-based integrity.

Chapter 41

Error Correction Codes

Hamming codes, Reed-Solomon codes, LDPC codes, convolutional codes.

Chapter 42

Cyclic Redundancy Checks (CRC)

CRC polynomials, CRC-8, CRC-16, CRC-32 standards, efficient implementation using tables, applications in data integrity.

Chapter 43

Data Serialization Fundamentals

Converting in-memory structures to byte streams, portability issues.

Chapter 44

Binary Serialization Formats

Protocol Buffers, FlatBuffers, Cap'n Proto, MessagePack, CBOR.

Chapter 45

Text Serialization Formats

JSON, XML, YAML, TOML, human-readable vs. machine-efficient tradeoffs.

Chapter 46

Custom Binary Protocols

Designing efficient binary formats, alignment considerations, versioning.

Chapter 47

Data Compression Theory

Information theory basics, entropy, lossless vs. lossy bounds.

Chapter 48

Dictionary-Based Compression

LZ77, LZ78, LZW, Lempel-Ziv family, dictionary construction.

Chapter 49

Entropy Coding

Huffman coding, arithmetic coding, asymmetric numeral systems (ANS).

Chapter 50

Modern Compression Algorithms

zlib, gzip, bzip2, LZMA, Zstandard, Brotli, compression levels.

Chapter 51

Specialized Compression

Run-length encoding, delta encoding, columnar compression, domain-specific methods.

Part II

Computer Architecture & Logic

ARRAYS EXIST *in physical hardware. To understand array performance, we must understand the machines that execute our code—from transistors to complete systems. This part builds a complete picture of computational machinery.*

What Makes This Different:

- ***Silicon to Software:*** Complete vertical integration
- ***Modern Architecture:*** Multi-core, SIMD, heterogeneous systems
- ***Performance Foundations:*** Why code runs fast or slow
- ***Hardware-Software Contract:*** What each layer assumes

“The purpose of computing is insight, not numbers.”

— RICHARD HAMMING

Chapter 52

Semiconductor Physics Foundations

Silicon properties, doping (n-type, p-type), P-N junctions, depletion regions, how semiconductors enable switching.

Chapter 53

MOSFET Transistors

Gate, source, drain, channel, threshold voltage, NMOS and PMOS, how transistors act as switches.

Chapter 54

CMOS Logic Fundamentals

Complementary MOS, pull-up and pull-down networks, static power vs. dynamic power, why CMOS won.

Chapter 55

Logic Gate Implementation

CMOS implementation of NOT, NAND, NOR gates, gate delay, rise/fall time, fan-out limitations.

Chapter 56

Logic Gate Families

AND, OR, XOR, XNOR from NAND/NOR, transmission gates, tri-state buffers.

Chapter 57

Combinational Circuit Design

Truth tables to logic expressions, Karnaugh maps, minimization, multi-level logic.

Chapter 58

Arithmetic Circuits: Adders

Half adder, full adder, ripple-carry adder, carry-lookahead adder, carry-select adder.

Chapter 59

Arithmetic Circuits: Multipliers and Dividers

Array multiplier, Booth multiplier, Wallace tree, division algorithms, restoring vs. non-restoring.

Chapter 60

Multiplexers and Demultiplexers

2:1 mux, 4:1 mux, tree structures, using muxes for logic implementation.

Chapter 61

Encoders and Decoders

Binary encoding, priority encoding, 7-segment decoder, address decoding.

Chapter 62

Arithmetic Logic Circuits

Half adder, full adder, ripple-carry adder, carry-lookahead adder, carry-save adder, subtraction using two's complement, magnitude comparators.

Chapter 63

Advanced Arithmetic Units

Multipliers (array multiplier, Booth's algorithm, Wallace tree), dividers, floating-point units (FPUs), fused multiply-add (FMA).

Chapter 64

Sequential Logic Fundamentals

Difference between combinational and sequential logic, feedback, state, clock signals, synchronous versus asynchronous design.

Chapter 65

Comparators and Magnitude Detection

Equality comparator, magnitude comparator, signed vs. unsigned comparison.

Chapter 66

Sequential Logic Fundamentals

Need for memory elements, clock signals, synchronous vs. asynchronous design.

Chapter 67

Latches: SR, D Latches

Set-reset latch, gated D latch, transparent latch, latch vs. flip-flop distinction.

Chapter 68

Flip-Flops: D, JK, T Flip-Flops

Edge-triggered flip-flops, master-slave configuration, setup and hold time, clock-to-Q delay.

Chapter 69

Registers and Register Files

Parallel-load register, shift registers (SISO, SIPO, PISO, PIPO), register banks for CPU.

Chapter 70

Counters and Timers

Ripple counter, synchronous counter, up/down counter, modulo-N counter, timer circuits.

Chapter 71

Finite State Machines

Moore vs. Mealy machines, state diagrams, state table, state minimization, FSM implementation.

Chapter 72

Memory Cell Design: SRAM

6-transistor SRAM cell, read and write operations, stability analysis, SRAM array organization.

Chapter 73

Memory Cell Design: DRAM

1-transistor 1-capacitor cell, charge storage, sense amplifiers, refresh requirement.

Chapter 74

Memory Array Organization

Row and column decoding, wordline and bitline, memory cell array structure, address multiplexing.

Chapter 75

DRAM Operation and Timing

Row activation, column access, precharge, timing parameters (t_{RCD} , t_{RP} , t_{RAS} , CAS latency).

Chapter 76

DRAM Generations

SDR SDRAM, DDR, DDR2, DDR3, DDR4, DDR5, bandwidth evolution, power efficiency.

Chapter 77

Advanced DRAM Features

Burst mode, bank interleaving, dual-channel, command queuing, on-die termination.

Chapter 78

Non-Volatile Memory Technologies

Flash memory (NAND, NOR), program/erase cycles, wear leveling, EEPROM, ROM variants.

Chapter 79

Flash Memory Operations

Programming (writing), erasing, read operations, wear leveling, write amplification, garbage collection, over-provisioning.

Chapter 80

Emerging Memory Technologies

Phase-change memory (PCM), resistive RAM (ReRAM), magnetoresistive RAM (MRAM), 3D XPoint.

Chapter 81

Memory Hierarchy Fundamentals

Pyramid structure, latency-capacity tradeoffs, why hierarchies exist, locality principles.

Chapter 82

Cache Memory Fundamentals

Temporal locality, spatial locality, cache line (block), tag-data organization.

Chapter 83

Cache Mapping Strategies

Direct-mapped cache, fully-associative cache, set-associative cache (2-way, 4-way, 8-way, N-way).

Chapter 84

Cache Replacement Policies

LRU (least recently used), LFU, FIFO, random, pseudo-LRU, practical implementations.

Chapter 85

Cache Write Policies

Write-through, write-back, write-allocate, no-write-allocate, dirty bits.

Chapter 86

Multi-Level Cache Hierarchies

L1 instruction and data caches, unified L2 cache, L3 shared cache, inclusive vs. exclusive caches.

Chapter 87

Victim Caches and Advanced Structures

Victim cache for conflict misses, stream buffers for prefetching, trace caches.

Chapter 88

Cache Coherence Problem

Shared memory multiprocessors, cache inconsistency, coherence vs. consistency.

Chapter 89

Cache Coherence Protocols: MESI

Modified, Exclusive, Shared, Invalid states, state transitions, bus snooping.

Chapter 90

Cache Coherence Protocols: MOESI and Beyond

Owned state, directory-based coherence, scalability issues.

Chapter 91

Cache Performance Analysis

Hit rate, miss rate, average memory access time (AMAT), miss penalty, classifying misses (compulsory, capacity, conflict).

Chapter 92

Cache Optimization Techniques

Blocking/tiling, array padding, loop interchange, prefetching, cache-aware algorithms.

Chapter 93

Instruction Set Architecture (ISA)

RISC versus CISC philosophy, instruction encoding and formats, addressing modes, register architecture, condition codes.

Chapter 94

x86 Architecture

x86 instruction encoding, variable-length instructions, complex addressing modes, backward compatibility, x86-64 extensions.

Chapter 95

ARM Architecture

Load-store architecture, fixed-length instructions (ARM mode), Thumb instruction set, NEON SIMD, ARM versus x86 comparison.

Chapter 96

RISC-V Architecture

Open ISA, modular extensions, base integer instruction set, standard extensions (M, A, F, D, C), design philosophy.

Chapter 97

Pipelining Fundamentals

Instruction pipeline stages (IF, ID, EX, MEM, WB), throughput improvement, latency, ideal CPI, pipeline diagrams.

Chapter 98

Pipeline Hazards

Structural hazards, data hazards (RAW, WAR, WAW), control hazards, stalls and bubbles, hazard detection and resolution.

Chapter 99

Data Forwarding and Bypassing

Forwarding paths, bypassing results from later stages, forwarding logic, load-use hazard, limitations of forwarding.

Chapter 100

Branch Handling in Pipelines

Branch penalty, delayed branches, branch prediction necessity, flushing pipeline on misprediction, branch delay slots.

Chapter 101

Branch Prediction Techniques

Static prediction (always taken, always not-taken, backward taken forward not-taken), dynamic prediction, branch history.

Chapter 102

Advanced Branch Prediction

Two-bit saturating counters, branch history table (BHT), branch target buffer (BTB), two-level adaptive predictors, tournament predictors.

Chapter 103

Return Address Stack

Function call prediction, hardware return address stack, depth and accuracy, correlation with call-return patterns.

Chapter 104

Superscalar Architecture

Multiple issue, instruction-level parallelism (ILP), dispatch width, execution units, dependency checking, instruction window.

Chapter 105

Out-of-Order Execution

Tomasulo's algorithm, reservation stations, register renaming, reorder buffer (ROB), commit stage, speculative execution.

Chapter 106

Register Renaming

Eliminating false dependencies (WAR, WAW), physical versus architectural registers, register alias table (RAT), freelist management.

Chapter 107

Speculative Execution

Branch speculation, memory speculation, exceptions in speculative execution, precise interrupts, ROB and commit.

Chapter 108

Speculative Execution Vulnerabilities

Spectre attacks (branch target injection, bounds check bypass), Meltdown (rogue data cache load), side-channel exploitation, mitigations.

Chapter 109

Memory Hierarchy Principles

Principle of locality (temporal and spatial), access time gaps, capacity and cost tradeoffs, memory wall problem.

Chapter 110

Virtual Memory Fundamentals

Virtual address space, physical address space, address translation, memory protection.

Chapter 11

Page Tables and Address Translation

Page table entry structure, multi-level page tables, page table walk.

Chapter 112

Translation Lookahead Buffer (TLB)

TLB structure, TLB hit/miss, TLB reach, large pages (huge pages, superpages).

Chapter 113

Page Replacement Algorithms

Optimal (Belady), FIFO, LRU, clock (second chance), working set model.

Chapter 114

Demand Paging and Page Faults

Page fault handling, major vs. minor faults, swap space, thrashing.

Chapter 115

Memory Management Unit (MMU)

Hardware implementation, page table base register, protection bits, privilege levels.

Chapter 116

Memory Protection and Isolation

Process isolation, kernel vs. user space, protection rings, segmentation.

Chapter 117

CPU Microarchitecture Overview

Instruction fetch, decode, execute, memory, writeback, datapath and control path.

Chapter 118

Instruction Set Architecture Principles

Instruction formats, operand addressing, register vs. memory operands, RISC philosophy.

Chapter 119

CISC vs. RISC

Complex instruction set (x86), reduced instruction set (ARM, RISC-V), microcode, instruction encoding density.

Chapter 120

Register Architecture

General-purpose registers, special-purpose registers (PC, SP, flags), register windows, register renaming.

Chapter 121

Addressing Modes

Immediate, register direct, memory direct, register indirect, indexed, PC-relative, stack addressing.

Chapter 122

Instruction Encoding

Fixed-length vs. variable-length, opcode, operands, prefix bytes (x86), instruction alignment.

Chapter 123

Pipelining Fundamentals

Instruction pipeline stages, throughput vs. latency, ideal speedup, pipeline registers.

Chapter 124

Pipeline Hazards

Structural hazards, data hazards (RAW, WAR, WAW), control hazards (branches).

Chapter 125

Data Forwarding and Bypassing

Forwarding paths, resolving data hazards, bypass network.

Chapter 126

Pipeline Stalls and Bubbles

When forwarding isn't enough, NOP insertion, performance impact.

Chapter 127

Branch Prediction Fundamentals

Static prediction, dynamic prediction, branch history, importance for performance.

Chapter 128

Branch Prediction: Simple Schemes

1-bit predictor, 2-bit saturating counter, bimodal predictor.

Chapter 129

Branch Prediction: Advanced Schemes

Two-level adaptive predictors, local vs. global history, correlating predictors.

Chapter 130

Branch Target Buffer and Return Stack

BTB for target prediction, return address stack (RAS) for function returns, indirect branch prediction.

Chapter 131

Speculative Execution

Executing before knowing if needed, branch prediction enabling speculation, squashing mispredicted paths.

Chapter 132

Out-of-Order Execution

Instruction reordering, Tomasulo's algorithm, reservation stations, reorder buffer.

Chapter 133

Register Renaming

Eliminating false dependencies (WAR, WAW), physical vs. architectural registers, register allocation table.

Chapter 134

Instruction Scheduling

Issue width, instruction window, dependency checking, wake-up and select logic.

Chapter 135

Superscalar Execution

Multiple issue, dispatch width, execution units, commit width, IPC (instructions per cycle).

Chapter 136

Very Long Instruction Word (VLIW)

Static scheduling by compiler, explicit parallelism, no dependency checking hardware, Itanium example.

Chapter 137

Simultaneous Multithreading (SMT)

Hyper-threading, thread-level parallelism, sharing execution resources, Intel and AMD implementations.

Chapter 138

Memory Ordering and Consistency

Load/store reordering, memory fences, barriers, acquire-release semantics.

Chapter 139

Consistency Models

Sequential consistency, total store order (TSO), weak ordering, relaxed models.

Chapter 140

Input/Output Systems

I/O devices and controllers, I/O address space (port-mapped versus memory-mapped I/O), programmed I/O, interrupt-driven I/O.

Chapter 141

Interrupt Handling

Interrupt request (IRQ), interrupt vector table, interrupt service routine (ISR), interrupt priority, nested interrupts, interrupt latency.

Chapter 142

Direct Memory Access (DMA)

DMA controller, bus mastering, DMA transfer modes (burst, cycle stealing), scatter-gather DMA, reducing CPU overhead.

Chapter 143

I/O Buses and Interconnects

System bus, front-side bus, peripheral buses, PCIe (lanes, endpoints, root complex), bus arbitration, bandwidth and latency.

Chapter 144

Storage Controllers and Interfaces

SATA, SAS, NVMe, NVMe over Fabrics, command queuing, storage protocol stack, queue depth and parallelism.

Chapter 145

Multi-Core Processors

Chip multiprocessors (CMP), symmetric multiprocessing (SMP), shared L2/L3 caches, on-chip interconnect (ring, mesh, crossbar).

Chapter 146

NUMA Architecture

Non-uniform memory access, memory affinity, local versus remote memory, NUMA domains, NUMA-aware programming.

Chapter 147

SIMD and Vector Processing

Single instruction multiple data, vector registers, lane-based execution, packed operations.

Chapter 148

x86 SIMD Extensions

MMX, SSE, SSE2-SSE4, AVX, AVX2, AVX-512, register width evolution.

Chapter 149

ARM NEON and SVE

NEON instruction set, Scalable Vector Extension, predication, variable vector length.

Chapter 150

GPU Architecture Overview

Massively parallel architecture, streaming multiprocessors, SIMT execution model.

Chapter 151

GPU Execution Model

Threads, warps/wavefronts, thread blocks, grid, kernel launch, occupancy.

Chapter 152

GPU Memory Hierarchy

Global memory, shared memory, registers, constant memory, texture memory, L1/L2 caches.

Chapter 153

GPU Programming Models

CUDA, OpenCL, compute shaders, host-device interaction, kernel code.

Chapter 154

Input/Output Architecture

I/O devices, device controllers, I/O buses, programmed I/O vs. interrupt-driven I/O.

Chapter 155

Interrupt Mechanism

Interrupt vector table, interrupt service routines, interrupt priority, nested interrupts.

Chapter 156

Interrupt Handling Details

Context saving/restoration, interrupt latency, interrupt masking, edge vs. level triggered.

Chapter 157

Exception Handling

Faults, traps, aborts, exception vectors, privilege level transitions.

Chapter 158

Direct Memory Access (DMA)

DMA controllers, bus mastering, scatter-gather DMA, DMA descriptors, reducing CPU overhead.

Chapter 159

I/O Memory Management Unit (IOMMU)

Device address translation, DMA protection, virtualization support, IOMMU page tables.

Chapter 160

Bus Architecture and Protocols

System bus, memory bus, I/O bus, bus arbitration, split transactions.

Chapter 161

Interconnect Fabrics

Point-to-point links, crossbar switches, mesh networks, ring interconnects.

Chapter 162

PCIe Architecture

Lanes, link width, generations (1.0-6.0), packets and transactions, TLP/DLLP/Physical layers.

Chapter 163

Cache-Coherent Interconnects

Intel QPI/UPI, AMD Infinity Fabric, coherence traffic, non-uniform cache access (NUCA).

Chapter 164

Network-on-Chip (NoC)

On-chip routing, wormhole routing, virtual channels, deadlock avoidance.

Chapter 165

Power Management Mechanisms

Dynamic voltage and frequency scaling (DVFS), clock gating, power gating, C-states, P-states.

Chapter 166

Thermal Design and Throttling

Thermal design power (TDP), temperature monitoring, thermal throttling, cooling solutions.

Chapter 167

Hardware Performance Counters

Performance monitoring units (PMU), counting events (cycles, instructions, cache misses, branch mispredictions).

Chapter 168

Profiling with Hardware Counters

Perf tools, VTune, hardware event sampling, attributing performance to code.

Chapter 169

Hardware Security Features

Secure boot, trusted execution environments (SGX, TrustZone), memory encryption (SEV, TME).

Chapter 170

Spectre and Meltdown

Speculative execution vulnerabilities, side-channel attacks, mitigations (KPTI, retpoline, microcode updates).

Chapter 171

Memory Safety Mechanisms

Bounds checking (Intel MPX), pointer authentication (ARM PAC), tagged memory, stack canaries.

Chapter 172

Hardware Debugging Infrastructure

JTAG, boundary scan, trace debugging, watchpoints, breakpoints.

Chapter 173

Mobile Device System-on-Chip (SoC) Architecture

Integrated CPU, GPU, modem, ISP on single die, ARM Cortex cores, heterogeneous computing, power islands, mobile SoC vendors (Qualcomm, Apple, MediaTek).

Chapter 174

Mobile Power Management IC (PMIC)

Voltage regulation, power sequencing, buck converters, LDO regulators, battery charging control, thermal management, power state transitions.

Chapter 175

Mobile Display Technology and Controllers

LCD vs OLED, display driver IC, timing controller (TCON), MIPI DSI interface, backlight control, touch controller integration, digitizer architecture.

Chapter 176

Mobile Battery Management Systems

Lithium-ion chemistry, charge cycles, protection circuits, fuel gauge IC, State of Charge (SoC) estimation, State of Health (SoH), battery authentication.

Chapter 177

Mobile Charging Circuit Architecture

USB Power Delivery, Quick Charge, voltage/current regulation, thermal throttling, charge pump topology, wireless charging (Qi standard), reverse charging.

Chapter 178

Mobile Baseband and RF Subsystems

Modem architecture, RF transceiver, power amplifier, antenna tuning, MIMO configuration, cellular bands, signal integrity, EMI considerations.

Chapter 179

Mobile Audio Codec and Amplifier

Digital-to-analog conversion, headphone amplifier, speaker amplifier, noise cancellation, audio routing, microphone bias, impedance matching.

Chapter 180

Mobile Camera ISP and Sensor Interface

Image signal processor, MIPI CSI, sensor array, autofocus driver, optical image stabilization (OIS), electronic image stabilization (EIS).

Chapter 181

Motherboard Component Identification

Visual inspection techniques, component labeling conventions, voltage regulator modules (VRM), chipset identification, power delivery phases, capacitor types (electrolytic, ceramic, tantalum).

Chapter 182

Desktop Power Supply Unit Architecture

ATX standard, 12V rails, 5V standby, power good signal, overcurrent protection, ripple suppression, efficiency ratings (80 Plus), modular vs non-modular.

Chapter 183

Power Supply Diagnostic Techniques

Paperclip test, multimeter voltage verification, load testing, ripple measurement, fan operation check, capacitor bulging inspection, burning smell identification.

Chapter 184

RAM Architecture and Organization

DIMM vs SO-DIMM, DDR generations, module organization (single-rank, dual-rank), SPD (Serial Presence Detect), XMP profiles, memory timings (CL, tRCD, tRP, tRAS).

Chapter 185

RAM Testing Methodologies

MemTest86, Windows Memory Diagnostic, POST beep codes, slot isolation testing, voltage verification, compatibility checking, ECC error detection.

Chapter 186

Storage Controller Interfaces

SATA controller, AHCI vs RAID mode, NVMe controller, M.2 key types (B, M, B+M), PCIe lane allocation, hot-swap capability, TRIM command.

Chapter 187

Hard Drive Diagnostic Tools

SMART attributes (reallocated sectors, pending sectors, CRC errors), surface scan, bad sector mapping, firmware corruption, mechanical failure sounds (clicking, grinding).

Chapter 188

SSD Architecture and Diagnostics

NAND flash types (SLC, MLC, TLC, QLC), wear leveling, garbage collection, over-provisioning, TRIM effectiveness, SSD-specific SMART attributes, controller firmware updates.

Chapter 189

Thermal Management System Analysis

Heat sink design, thermal paste application, thermal pads, heat pipe operation, vapor chamber cooling, thermal interface material (TIM) degradation, temperature monitoring (thermal diodes, thermistors).

Chapter 190

Voltage Regulation Module (VRM) Operation

Buck converter topology, PWM controller, MOSFET switching, inductor selection, output capacitor filtering, load line calibration, voltage droop, VRM phase count.

Chapter 191

Component-Level Voltage Testing

Reference voltage points, power rail sequencing, voltage ripple measurement, load-induced voltage drop, ground plane continuity, short circuit detection.

Chapter 192

Circuit Board Visual Inspection

Burnt components identification, electrolytic capacitor failure (bulging, leaking), cold solder joints, cracked traces, corrosion detection, flux residue, water damage indicators.

Chapter 193

Multimeter Diagnostic Techniques

Continuity testing, diode testing mode, capacitor ESR estimation, resistance measurement interpretation, AC/DC voltage verification, current measurement safety.

Chapter 194

Oscilloscope Diagnostics for Repair

Signal integrity verification, clock signal analysis, power supply ripple measurement, PWM duty cycle measurement, trigger setup, probe compensation, bandwidth requirements.

Chapter 195

Logic Analyzer for Bus Debugging

I2C bus analysis, SPI communication verification, USB signal quality, protocol decoding, timing violation detection, multi-channel capture.

Chapter 196

Component Identification and Datasheet Reading

SMD marking codes, component package types (SOT, SOIC, QFN, BGA), pin 1 identification, polarity markings, datasheet interpretation (absolute maximum ratings, typical operating conditions).

Chapter 197

Schematic and PCB Layout Reading

Schematic symbol conventions, net naming, component reference designators, power and ground symbols, PCB layer stackup, trace routing, via types, copper pour, silkscreen interpretation.

Chapter 198

SMD Soldering and Desoldering

Soldering iron temperature control, flux application, component alignment, heat dissipation management, hot air rework station usage, desoldering braid, solder sucker.

Chapter 199

BGA and QFN Rework Techniques

Ball grid array structure, reflow profile (preheat, soak, reflow, cooling), X-ray inspection, underfill removal, stencil alignment, solder ball placement.

Chapter 200

Microsoldering Under Microscope

Magnification selection, lighting techniques, steady hand positioning, tweezers usage, fine-pitch component handling, flux pen application, solder wire diameter selection.

Chapter 201

Component Replacement Decision Matrix

Cost-benefit analysis, availability considerations, compatibility verification, firmware dependencies, security chip replacement implications, serialized component handling.

Chapter 202

ESD Protection During Repair

Electrostatic discharge fundamentals, wrist strap usage, ESD mat grounding, ionizer benefits, component handling procedures, humidity control, triboelectric series.

Chapter 203

Diagnostic Flowchart Construction

Symptom identification, component isolation, probability-based testing order, decision tree optimization, documentation importance, feedback loop integration.

Chapter 204

Firmware and BIOS Diagnostics

BIOS POST codes, UEFI firmware interface, CMOS battery replacement, BIOS recovery modes, firmware corruption detection, update procedures, downgrade considerations.

Chapter 205

Power-On Self-Test (POST) Analysis

POST sequence stages, beep code interpretation by manufacturer, LED diagnostic codes, minimal POST configuration, component isolation testing, POST card usage.

Chapter 206

No Power Diagnosis Methodology

Power button continuity, PSU verification, motherboard standby voltage, short circuit detection, component-by-component isolation, breadboard testing.

Chapter 207

No Display Troubleshooting

Monitor vs system isolation, GPU reseating, CMOS reset, RAM testing, integrated vs discrete graphics, display cable verification, external display testing.

Chapter 208

Intermittent Fault Diagnosis

Temperature-dependent failures, connector oxidation, solder joint cracking, capacitor degradation, environmental factor correlation, stress testing.

Chapter 209

Water Damage Recovery Procedures

Immediate power disconnection, disassembly urgency, ultrasonic cleaning, isopropyl alcohol washing, corrosion neutralization, drying time requirements, component-level inspection.

Chapter 210

Board-Level Troubleshooting Strategy

Power rail verification sequence, signal path tracing, component substitution testing, known-good board comparison, documentation review, community knowledge utilization.

Chapter 211

Mobile Device Disassembly Techniques

Adhesive removal (heat gun, isopropyl alcohol), spudger usage, screw organization, cable connector types (ZIF, FPC, BTB), shielding can removal, warranty seal implications.

Chapter 212

Mobile Display Assembly Replacement

Digitizer vs LCD separation, frame transfer, adhesive application, cable routing, testing before full assembly, touch calibration, backlight testing.

Chapter 213

Mobile Battery Replacement Procedures

Adhesive strip removal, pull tab technique, battery connector identification, swelling detection, capacity verification, recycling considerations, calibration requirements.

Chapter 214

Mobile Charging Port Replacement

Daughterboard identification, flex cable replacement, direct mainboard soldering, pin alignment verification, USB-C orientation testing, data line continuity.

Chapter 215

Mobile IC-Level Repair Techniques

Hotplate vs hot air station selection, flux type for rework, underfill challenges, shield can handling, thermal coupling considerations, component height matching.

Chapter 216

Mobile Baseband Repair and IMEI

Baseband processor role, IMEI storage locations, legal considerations, carrier lock implications, baseband firmware flashing, signal strength testing.

Chapter 217

Mobile Microphone and Speaker Replacement

Component location identification, adhesive vs clip mounting, polarity verification, sound quality testing, water damage impact, mesh cleaning.

Chapter 218

Desktop Motherboard Diagnostic LEDs

Debug LED interpretation, CPU, RAM, VGA, BOOT sequence indicators, Q-code displays, troubleshooting based on LED patterns.

Chapter 219

CPU Installation and Socket Inspection

LGA vs PGA socket types, pin damage assessment, bent pin straightening techniques, thermal paste application patterns, mounting pressure, cooler compatibility.

Chapter 220

GPU Troubleshooting and Repair

Artifacting causes, memory chip failure, VRM failure, reflow as temporary fix, VRAM testing, power connector verification, driver vs hardware isolation.

Chapter 221

Storage Device Recovery Techniques

Logical vs physical failure, partition table corruption, filesystem repair, firmware corruption, PCB swap considerations, data recovery services, clean room requirements.

Chapter 222

Laptop-Specific Repair Challenges

Proprietary components, serviceability variation, keyboard replacement, screen hinge repair, DC jack replacement, trackpad connector issues, cooling system maintenance.

Chapter 223

Network Interface Diagnostics

Ethernet PHY testing, link light interpretation, loopback testing, MAC address verification, speed negotiation issues, cable tester usage.

Chapter 224

USB Port Repair and Diagnosis

USB protocol layers (physical, link, transaction), port power issues, data line integrity, device enumeration failure, controller chip identification, per-port fuse/limiting.

Chapter 225

Audio System Troubleshooting

Codec chip identification, HD Audio vs AC'97, front panel header pinout, impedance sensing, pop/click diagnosis, amplifier testing, line-out vs headphone-out.

Chapter 226

Thermal Imaging for Fault Detection

Infrared camera operation, thermal hotspot identification, component overheating patterns, short circuit localization, VRM phase imbalance detection.

Chapter 227

Power Supply Load Testing

Electronic load usage, dynamic load testing, cross-load testing, hold-up time measurement, transient response testing, aging effects.

Chapter 228

Signal Integrity Testing Equipment

Time-domain reflectometry (TDR), vector network analyzer (VNA), bit error rate testing (BERT), eye diagram analysis, jitter measurement.

Chapter 229

X-Ray Inspection for BGA Repair

X-ray system operation, void detection in solder balls, bridge identification, component alignment verification, non-destructive testing benefits.

Chapter 230

Micro-Rework Station Setup

Hot air station temperature profiling, preheater usage, fume extraction, ESD-safe workspace, tool organization, component storage, microscope positioning.

Chapter 231

Repair Documentation Best Practices

Photography during disassembly, screw mapping, cable routing documentation, settings preservation (BIOS, driver versions), customer communication, warranty tracking.

Chapter 232

Component Sourcing and Authenticity

Genuine vs counterfeit parts, supplier reliability, date code interpretation, packaging inspection, electrical testing for verification, gray market risks.

Chapter 233

Repair Cost-Benefit Analysis

Labor time estimation, parts cost, tool amortization, success probability, data recovery value, replacement cost comparison, customer budget consideration.

Chapter 234

Safety Protocols in Electronics Repair

Capacitor discharge procedures, lithium battery handling, lead-free vs leaded solder, ventilation requirements, fire safety, chemical handling (flux, cleaning agents).

Chapter 235

Professional Tool Selection for Repair

Investment priority, quality vs cost tradeoff, calibration requirements, maintenance needs, ergonomics, expandability, brand comparison.

Chapter 236

Repair Business Quality Assurance

Post-repair testing checklist, burn-in testing duration, stress testing tools, warranty period determination, failure rate tracking, customer satisfaction metrics.

Chapter 237

Retro Hardware Repair Techniques

Capacitor replacement (recapping), tantalum capacitor substitution, BIOS battery replacement, floppy drive alignment, CRT monitor safety, vintage connector types.

Chapter 238

Industrial and Embedded System Repair

Extended temperature component identification, conformal coating removal, ruggedized hardware, industrial power supplies, legacy protocol support, long-term availability.

Chapter 239

Server Hardware Diagnostics

IPMI and BMC functionality, out-of-band management, SEL (System Event Log) analysis, predictive failure analysis, redundant component testing, hot-swap procedures.

Chapter 240

Repair Training and Skill Development

Deliberate practice strategies, sacrificial board usage, component identification games, soldering certification programs, online community resources, mentorship importance.

Chapter 241

Design for Testability

Scan chains, built-in self-test (BIST), fault injection, manufacturing test.

Chapter 242

Hardware Description Languages

Verilog and SystemVerilog, VHDL, behavioral vs. structural modeling, simulation vs. synthesis.

Chapter 243

Digital Design Flow

RTL design, logic synthesis, place and route, timing closure, physical design.

Chapter 244

FPGA Architecture

Look-up tables (LUTs), configurable logic blocks (CLBs), programmable interconnect, block RAM, DSP slices.

Chapter 245

ASIC vs. FPGA Tradeoffs

Performance, power, cost, flexibility, development time, reconfigurability.

Glossary

Reflections at the End

As you turn the final pages of **Arliz**, I invite you to pause—just for a moment—and look back. Think about the path you’ve taken through these chapters. Let yourself ask:

“Wait... what just happened? What did I actually learn?”

I won’t pretend to answer that for you. The truth is—****only you can****. Maybe it was a lot. Maybe it wasn’t what you expected. But if you’re here, reading this, something must have kept you going. That means something.

This book didn’t start with a grand plan. It started with a simple itch: **What even is an array, really?** What began as a curiosity about a “data structure” became something much stranger and—hopefully—much richer. We wandered through history, philosophy, mathematics, logic gates, and machine internals. We stared at ancient stones and modern memory layouts and tried to see the invisible threads connecting them.

If that sounds like a weird journey, well—yeah. It was.

This is Not the End

Arliz isn’t a closed book. It’s a snapshot. A frame in motion. And maybe your understanding is the same. You’ll return to these ideas later, years from now, and see new angles. You’ll say, “Oh. That’s what it meant.” That’s good. That’s growth.

Everything you’ve read here is connected to something bigger—algorithms, networks, languages, systems, even the people who built them. There’s no finish line. And that’s beautiful.

From Me to You

If this book gave you something—an idea, a shift in thinking, a pause to wonder—then it has done its job. If it made you feel like maybe programming isn’t just code and rules, but a window into something deeper—then that means everything to me.

Thank you for being here.
Thank you for not skipping the hard parts.
Thank you for choosing to think.

One More Thing

You're not alone in this.
The Arliz project lives on GitHub, and the conversations around it will (hopefully) continue. If you spot mistakes, have better explanations, or just want to say hi—come by. Teach me something. Teach someone else. That's the best way to say thanks. Knowledge grows in community.
So share. Build. Break. Rebuild.
Ask better questions.
And always, always—stay curious.

Final Words

Arrays were just the excuse.
Thinking was the goal.
And if you've started to think more clearly, more deeply, or more historically about what you're doing when you write code—then this wasn't just a technical book. It was a human one.
Welcome to the quiet, lifelong joy of understanding.

This completes the first living edition of Arliz.

Thank you for joining this journey from zero to arrays, from ancient counting to modern computation.

The exploration continues...

Author's Notes and Reflections

On Naming Conventions and Creative Processes

I should confess something about my naming process: I tend to pick names first and find meaningful justifications later. Very scientific, I know! The name "Arliz" started as a random choice that simply felt right phonetically. Only after committing to it did I discover the backronym that now defines its meaning. This probably says something about my creative process—intuition first, rationalization second.

This approach extends beyond naming. Many aspects of this book emerged organically from curiosity rather than systematic planning. What began as personal notes to understand arrays evolved into a comprehensive exploration of computational thinking itself.

On Perfectionism and Living Documents

You should know that many of the algorithms presented in this book are my own implementations, built from first principles rather than copied from optimized sources. This means you might encounter code that runs slower than industry standards—or occasionally faster, when serendipity strikes.

Some might view this as a weakness, but I consider it a feature. The goal isn't to provide the most optimized implementations but to demonstrate the thinking process that leads to understanding. When you can reconstruct a solution from fundamental principles, you've achieved something more valuable than memorizing an optimal algorithm.

On Academic Formality and Personal Voice

You might notice that this book alternates between formal academic language and more conversational tones. This is intentional. While I respect the precision that formal writing provides, I also believe that learning happens best in an atmosphere of intellectual friendship rather than academic intimidation.

When I suggest you could "use this book as a makeshift heating device" if you find the approach ridiculous, I'm not being flippant—I'm acknowledging that not every approach works for every learner. Intellectual honesty includes admitting when your methods might not suit your audience.

On Scope and Ambition

The scope of this book—from ancient counting to modern distributed systems—might seem overly ambitious. Some might argue that such breadth necessarily sacrifices depth. I disagree, but I understand the concern.

My experience suggests that understanding connections between disparate fields often provides insights that narrow specialization misses. When you see arrays as part of humanity's broader intellectual project, you understand them differently than when you see them as isolated programming constructs.

That said, if you find the historical sections tedious or irrelevant, you have my permission to skip ahead. The book is designed to be valuable even when read non-sequentially.

On Errors and Imperfection

I mentioned that you'll find errors in this book. This isn't false modesty—it's realistic acknowledgment. Complex explanations, mathematical derivations, and code implementations inevitably contain mistakes, especially in a work that grows and evolves over time.

Rather than viewing this as a flaw, I encourage you to see it as an opportunity for engagement. When you find an error, you're not just identifying a problem—you're participating in the process of building better understanding. The best learning often

happens when we encounter and resolve contradictions.

On Time Investment and Expectations

When I suggest this book requires months rather than weekends to master, I'm not trying to inflate its importance. Complex concepts genuinely require time to internalize. Mathematical intuition develops gradually, through repeated exposure and active practice.

If you're looking for quick solutions to immediate programming problems, this book will frustrate you. If you're interested in developing the kind of deep understanding that serves you throughout your career, the time investment will prove worthwhile.

On Community and Collaboration

This book exists because of community—the open-source community that provides tools and resources, the academic community that develops and refines concepts, and the programming community that applies these ideas in practice.

Your engagement with this material makes you part of that community. Whether you find errors, suggest improvements, or simply work through the exercises thoughtfully, you're contributing to the collective understanding that makes books like this possible.

Final Reflection

Writing this book has been an exercise in understanding my own learning process. I've discovered that I learn best by building connections between disparate ideas, by understanding historical context, and by implementing concepts from scratch rather than accepting them as given.

Your learning process might be entirely different. Use what serves you from this book, adapt what needs adaptation, and don't hesitate to supplement with other resources when my explanations fall short.

The goal isn't for you to learn exactly as I did, but for you to develop your own path to deep understanding.

These notes reflect thoughts and observations that didn't fit elsewhere but seemed worth preserving. They represent the informal side of a formal exploration—the human element in what might otherwise seem like purely technical content.